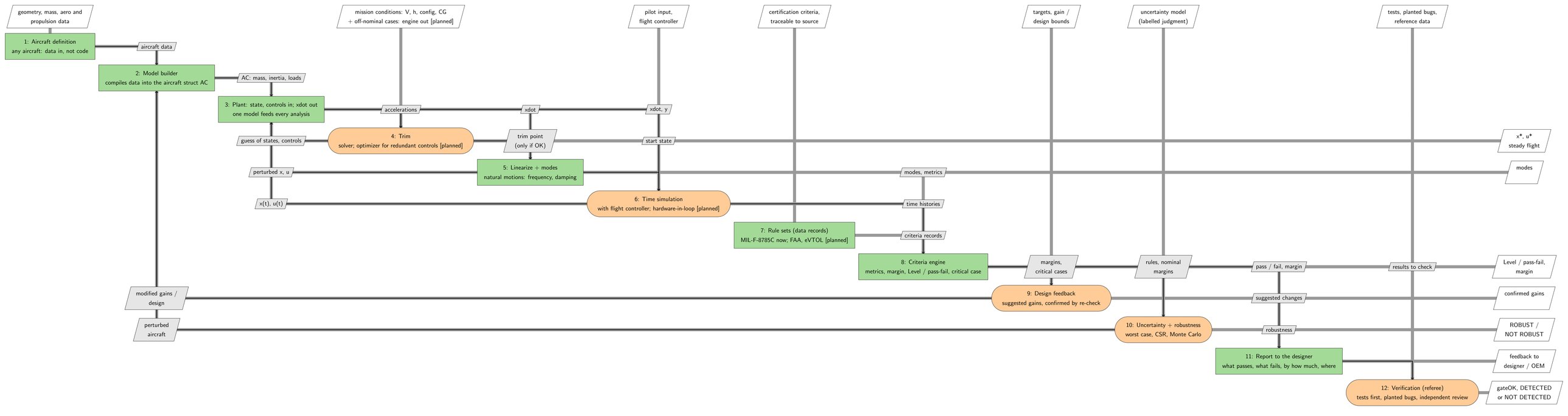
geometry, mass, aero and propulsion data
mission conditions: V, h, config, CG + off-nominal cases: engine out [planned]
pilot input, flight controller
certification criteria, traceable to source
targets, gain / design bounds
uncertainty model (labelled judgment)
tests, planted bugs, reference data
1: Aircraft definition any aircraft: data in, not code
aircraft data
2: Model builder compiles data into the aircraft struct AC
AC: mass, inertia, loads
3: Plant: state, controls in; xdot out one model feeds every analysis
accelerations
xdot
xdot, y
guess of states, controls
4: Trim solver; optimizer for redundant controls [planned]
trim point (only if OK)
start state
x*, u* steady flight
perturbed x, u
5: Linearize + modes natural motions: frequency, damping
modes, metrics
modes
x(t), u(t)
6: Time simulation with flight controller; hardware-in-loop [planned]
time histories
7: Rule sets (data records) MIL-F-8785C now; FAA, eVTOL [planned]
criteria records
8: Criteria engine metrics, margin, Level / pass-fail, critical case
margins, critical cases
rules, nominal margins
pass / fail, margin
results to check
Level / pass-fail, margin
modified gains / design
9: Design feedback suggested gains, confirmed by re-check
suggested changes
confirmed gains
perturbed aircraft
10: Uncertainty + robustness worst case, CSR, Monte Carlo
robustness
ROBUST / NOT ROBUST
11: Report to the designer what passes, what fails, by how much, where
feedback to designer / OEM
12: Verification (referee) tests first, planted bugs, independent review
gateOK, DETECTED or NOT DETECTED